Reviewing fundamental concepts in Python

Objects

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Module 01: Essential

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AGENDA

Sequence objects

- list
- tuple

Mapping objects

- dictionary

Built-in object

Object type	Example
lists	[1, [2, ‘three’], 3.5] list(range(10))
dictionaries	{‘food’:egg, ‘taste’:salty} dict(hours=24, year=365)
tuples	(1, ‘U’, 2, ‘union) tuple(‘spam’)
sets	set('abc') {'b', 'c', 'a'}

Class exercise

04

Talk ❢ Pair ❢ Share

What are some scenario that you would use a particular built-in object over another?

- List
- Dictionary
- Tuple
- Sets

Write your answers on the back of your name tag (with date, please)

Lists

- General sequence object
- Positionally ordered collections of arbitrarily typed objects
- Mutable

```
my_list = [123, ‘123’, 1.23, 12, 3]
```

Sequence operation

1. How to determine the number of items in the list?
2. How do you slice a list and return a new list?
3. How to concatenate to make a new list?
4. Did you change the original list?

Nested Lists

A list with nested list

```
>>> NL = [[1, 2, 3],
          [4, 5, 6],
          [7, 8, 9]]
```

A nested list that is a 3x3 matrix

Operations to fetch

1. Get the 2nd row ([4, 5, 6]).
2. Get the 3rd item in the 2nd row (6).

List comprehension expression

A way to build a new list by running an expression on each item in a sequence (from left to right).

```
>>> NL = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
>>> column2 = [row[1] for row in NL]
>>> column2
```

What will `column2` return?

Class exercise

Given the matrix

```
NL = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
```

CL 3.1:

Write a list comprehension to collect the diagonal items from matrix

Expected output [1, 5, 9]

Hint: Use a for loop to iterate through the list index

Sequence object: Tuple support mixed types and nesting

Immutable: Use to represent a fixed collection of items

```
T = (1, 3, 5, 7, 11)
```

Sequencing operation

1. Can you concatenate to a tuple?
2. Can you change a tuple?
3. How do you index a tuple?

Class exercise

```
my_list = [1, 2, 3, 4, 5]
```

CL 3.2:

Create a list of tuples where tuple is (n, n**3) where n is each number in the list.

Expected output [(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)]

Dictionaries

- Mapping object
- Map keys to associated values
- Indexing is not by relative position, rather by key
- Mutable

```
job = {‘name’: {‘first’:’John’, ‘last’:’Smith’},
       ‘jobs’: [‘dev’, ‘mgr’],
       ‘age’: 39}
```

Mapping operation

1. How do you index the nested dictionary?
2. How would you append to the job key?

Class exercise


```
num_dict = {'num1':100, 'num2':346, 'num3':-24}
```

CL3.3:

Multiple all the values of the items in a dictionary